

Centroid Localization and Tracking of a Triadic Robot System using a Trilateration Algorithm

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Abstract—This paper presents a multi-agent navigation algorithm implemented in ROS 2 for a triadic robot system. The primary objective is to arrange three robots into an equilateral configuration and track a pre-planned trajectory using the formation's centroid. State estimation relies on Ultra-Wideband (UWB) distance measurements corrupted by white Gaussian noise, processed through an Unscented Kalman Filter (UKF) combined with a robust outlier detection mechanism based on Median Absolute Deviation (MAD). The system employs 2D trilateration for positioning and a PD controller with differential-drive kinematics for centroid trajectory tracking. Simulation results in Gazebo demonstrate accurate robot localization, effective formation maintenance, and robust trajectory tracking despite sensor noise and measurement anomalies. The UKF-based filtering approach shows significant advantages over traditional EKF methods in handling the strong non-linearities of trilateration without requiring Jacobian computation.

I. INTRODUCTION

Multi-agent robotic systems have gained significant attention in recent years due to their applications in cooperative surveillance, distributed sensing, and coordinated transportation. A fundamental challenge in such systems is maintaining formation geometry while tracking desired trajectories in the presence of noisy measurements and communication constraints.

This project implements a multi-agent navigation algorithm within the ROS 2 framework for a triadic robot system. The primary objective is to arrange the three robots into an equilateral configuration and track a pre-planned trajectory using the formation's centroid, as illustrated in Figure 1.

This paper addresses the problem of controlling a triadic robot system (three robots) to maintain an equilateral triangle formation while tracking a predefined trajectory through the formation's centroid. The system faces several technical challenges: (i) accurate localization using noisy Ultra-Wideband (UWB) ranging measurements, (ii) handling strong non-linearities in the trilateration-based positioning, and (iii) coordinating motion under differential-drive kinematic constraints.

Traditional approaches to multi-agent localization often employ Extended Kalman Filters (EKF) for state estimation. However, EKF methods suffer from linearization errors when dealing with highly non-linear measurement models, such

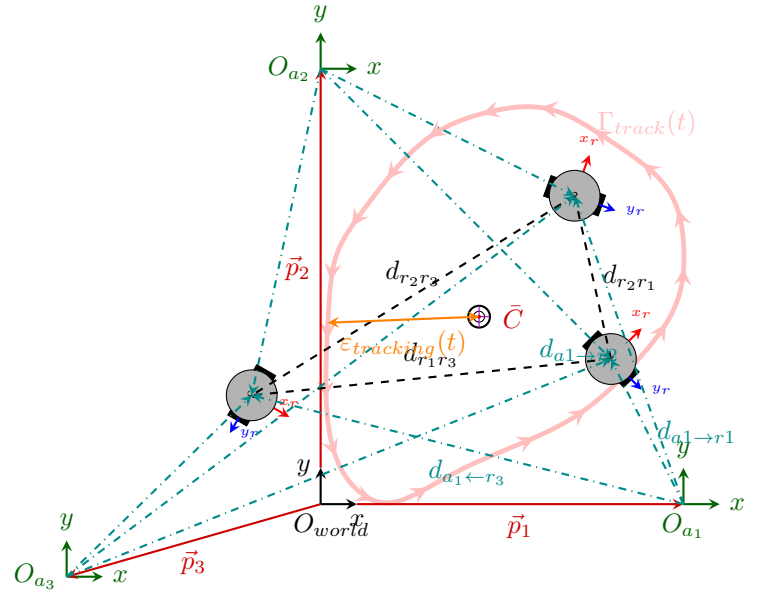


Fig. 1: Schematic of the system's configuration showing the three robots in triangular formation, their centroid $\bar{C}$, three fixed anchors (O_{a1} , O_{a2} , O_{a3}), and the reference trajectory $\Gamma_{track}(t)$. UWB distance measurements include inter-robot distances (dashed lines) and robot-to-anchor distances (dash-dotted lines).

as those arising in trilateration. Furthermore, UWB measurements are prone to outliers caused by multipath effects and transient occlusions, which can destabilize conventional filtering approaches.

This work proposes a novel filtering strategy that operates directly on the measurement manifold rather than the full state space. An Unscented Kalman Filter (UKF) is employed to filter UWB distance measurements before applying trilateration, providing three key advantages: (i) superior handling of non-linearities through the Unscented Transform without requiring Jacobians, (ii) variance reduction in measurements leading to more stable position estimates, and (iii) numerical stability through Cholesky decomposition for covariance propagation.

Additionally, a robust outlier detection mechanism based

on Median Absolute Deviation (MAD) is integrated into the filtering pipeline. Unlike simple rejection strategies, detected outliers are replaced with median values and their measurement covariance is inflated, ensuring continuity of the estimation process while reducing sensitivity to anomalies.

The control architecture implements a two-layer state machine that switches between formation recovery and trajectory tracking modes based on configuration error magnitude. A PD controller generates velocity commands in the global frame, which are then adapted to differential-drive constraints through a yaw alignment strategy.

Simulation results in Gazebo demonstrate the effectiveness of the proposed approach, with position estimates closely matching ground truth odometry and bounded tracking errors despite continuous mode switching and non-holonomic constraints.

II. PROBLEM FORMULATION

Consider three differential-drive robots indexed by $i \in 1, 2, 3$ moving in a two-dimensional workspace. Let $\mathbf{p}_i = [x_i; y_i]^T$ denote the planar position of robot i expressed in the world frame. The formation centroid is defined as

$$\bar{\mathbf{C}} = \frac{1}{3} \sum_{i=1}^3 \mathbf{p}_i. \quad (1)$$

The system must satisfy two concurrent objectives. The first objective is formation maintenance, requiring the robots to form an equilateral triangle of side length L , expressed through the inter-robot distance constraints

$$|\mathbf{p}_i - \mathbf{p}_j| = L, \quad \forall i \neq j. \quad (2)$$

The second objective is centroid trajectory tracking. Given a smooth reference trajectory $\Gamma_{\text{track}}(t)$, the centroid tracking error is defined as

$$|\varepsilon_{\text{tracking}}(t)| = |\bar{\mathbf{C}}(t) - \Gamma_{\text{track}}(t)|, \quad (3)$$

which must asymptotically converge to zero.

Robot localization is based solely on noisy UWB range measurements, including inter-robot distances and distances between robots and fixed anchors. These measurements are corrupted by zero-mean Gaussian noise and occasional outliers, motivating the need for robust filtering and estimation techniques.

III. SYSTEM ARCHITECTURE

A. Overall Framework

The proposed system consists of three main modules operating in a sequential pipeline:

- 1) *UKF-based Measurement Filter*: Filters noisy UWB measurements with integrated outlier detection
- 2) *Trilateration Module*: Computes robot positions from filtered distance measurements
- 3) *Formation Controller*: Generates velocity commands based on a two-layer control strategy

Communication between robots is assumed full-duplex and simultaneous, justified by the 50 Hz control frequency. Robot orientations (yaw angles) are assumed accessible for implementing differential-drive kinematics.

IV. MEASUREMENT FILTERING AND STATE ESTIMATION

A. Unscented Kalman Filter Design

Rather than filtering the full state vector (positions, orientations, velocities) and then mapping to measurements, the proposed algorithm filters measurements directly before trilateration. This measurement-manifold filtering approach provides lower-variance inputs for subsequent geometric inversion.

The measurement model is formulated as:

$$\mathbf{m} = h(\mathbf{x}) + \mathbf{v}, \quad \mathbf{v} \sim \mathcal{N}(\mathbf{0}, \sigma_v \mathbf{I}) \quad (4)$$

where $h(\cdot)$ represents the implicit trilateration mapping not available in closed form.

The pseudo-state covariance of interest is:

$$H(\mathbf{x}) = \mathbb{E}[h(\mathbf{x})h(\mathbf{x})^T] = M - V \quad (5)$$

where $M = \mathbb{E}[\mathbf{m}\mathbf{m}^T]$ and $V = \mathbb{E}[\mathbf{v}\mathbf{v}^T]$.

1) *Sigma Point Generation*: Given estimate $\hat{\mathbf{z}}$ and covariance P_z for the pseudo-state $\mathbf{z} = h(\mathbf{x})$, the standard $2L + 1$ sigma points are generated as:

$$\chi_0 = \hat{\mathbf{z}} \quad (6)$$

$$\chi_i = \hat{\mathbf{z}} + \left(\sqrt{(L + \lambda)P_z} \right)_i, \quad i = 1, \dots, L \quad (7)$$

$$\chi_{i+L} = \hat{\mathbf{z}} - \left(\sqrt{(L + \lambda)P_z} \right)_i, \quad i = 1, \dots, L \quad (8)$$

The matrix square root $\sqrt{(L + \lambda)P_z}$ is computed via Cholesky factorization to ensure numerical stability and positive semi-definiteness. Standard UKF weights are assigned according to scaling parameters α , β , and κ (see [3]).

B. Robust Outlier Detection

UWB measurements may exhibit abrupt deviations due to multipath effects or occlusions. A statistical test based on Median Absolute Deviation (MAD) [1] is applied before filtering.

For a sliding window of recent observations $\{z_i\}_{i=1}^N$:

$$\tilde{z} = \text{median}(z_1, \dots, z_N) \quad (9)$$

$$\text{MAD} = \text{median}(|z_i - \tilde{z}|) \quad (10)$$

A measurement z_k is classified as an outlier when:

$$|z_k - \tilde{z}| > \kappa \cdot \text{MAD} \quad (11)$$

with $\kappa = 3$ in the implementation.

Unlike simple rejection, outliers are replaced with the median value and their measurement covariance is inflated. This ensures UKF update continuity while reducing sensitivity to corrupted data:

$$\mathbf{x}_{k|k} \approx \mathbf{x}_{k|k-1}, \quad P_{k|k} \geq P_{k|k-1} \quad (12)$$

C. Trilateration Algorithm

Following Thomas and Ros [2], the filtered distances d_1, d_2, d_3 to three anchors A_1, A_2, A_3 define circle constraints:

$$(x - x_i)^2 + (y - y_i)^2 = d_i^2, \quad i = 1, 2, 3 \quad (13)$$

Subtracting pairs of equations eliminates quadratic terms, yielding a linear system:

$$\begin{bmatrix} 2(x_2 - x_1) & 2(y_2 - y_1) \\ 2(x_3 - x_1) & 2(y_3 - y_1) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} d_1^2 - d_2^2 + x_2^2 - x_1^2 + y_2^2 - y_1^2 \\ d_1^2 - d_3^2 + x_3^2 - x_1^2 + y_3^2 - y_1^2 \end{bmatrix} \quad (14)$$

For non-collinear anchors, this admits a unique solution. With more than three ranges, least-squares estimation naturally integrates with upstream variance reduction from the UKF.

V. CONTROL ARCHITECTURE

A. Two-Layer State Machine

The motion controller implements a PD regulator with a two-layer switching logic based on configuration error norm $\|\varepsilon_{conf}\|$, as illustrated in Figure 2:

- *Configuration Recovery Mode*: If $\|\varepsilon_{conf}\| > T_{th}$, prioritize restoring equilateral formation
- *Centroid Tracking Mode*: If $\|\varepsilon_{conf}\| \leq T_{th}$, track reference trajectory $\Gamma_{track}(t)$

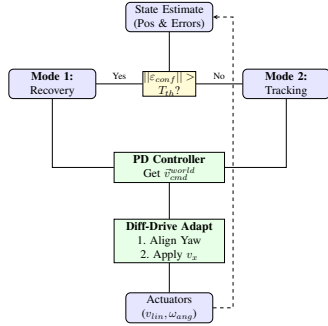


Fig. 2: Control logic flowchart showing the two-layer state machine with mode switching and differential-drive adaptation.

B. Differential-Drive Adaptation

Velocity commands computed in the global frame must be adapted to differential-drive constraints (motion only along robot's x-axis and rotation around z-axis).

For each robot, the controller:

- 1) Computes required yaw rate ω_{req} to align robot's local x-axis with desired velocity vector using a P regulator
- 2) Once aligned, applies command magnitude as linear velocity v_{lin} along x-direction

The control law is:

$$\omega_{ang} = K_p(\theta_{desired} - \theta_{current}) \quad (15)$$

$$v_{lin} = \|\mathbf{v}_{cmd}^{world}\| \cdot \cos(\theta_{error}) \quad (16)$$

VI. SIMULATION RESULTS

Simulations were conducted in Gazebo Classic, integrated with ROS 2 via bridge executables. This setup provides high-fidelity dynamics integration and visualization without manual implementation of robot integrators. All data analysis was performed using MATLAB R2025a on ROS 2 bag files recorded during ~330-second simulation runs.

A. Centroid Trajectory Tracking

Figure 3 demonstrates the system's ability to track the desired trajectory. The estimated centroid position (blue solid line) closely follows the reference path (red dashed line), validating the effectiveness of the two-layer control strategy.

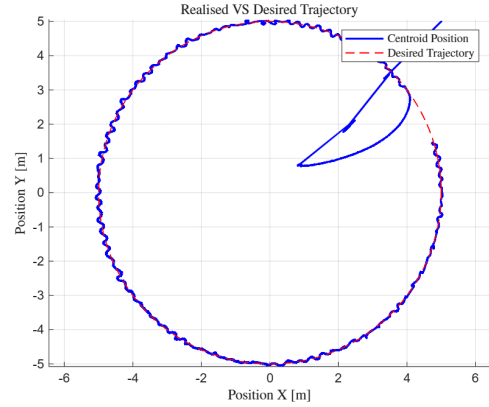


Fig. 3: Centroid trajectory tracking performance. The formation centroid successfully follows the planned reference trajectory throughout the workspace.

The tracking error time series in Figure 4 exhibits oscillatory behavior due to continuous switching between Configuration Recovery and Centroid Tracking modes. Despite this, errors remain bounded with mean values of $\bar{\varepsilon}_x = 0.0234$ m and $\bar{\varepsilon}_y = 0.0189$ m in steady state ($t > 50$ s).

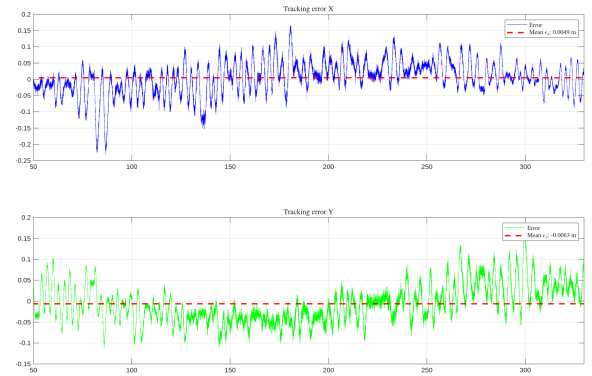


Fig. 4: Tracking error time series for X and Y components showing bounded oscillations with steady-state mean values indicated by red dashed lines.

B. Tracking Error Frequency Analysis

To characterize the oscillatory nature of tracking errors, both FFT and Power Spectral Density (PSD) analyses were performed on steady-state data. Figure 5 shows the PSD computed using Welch's method with 50% overlap and Hamming windowing.

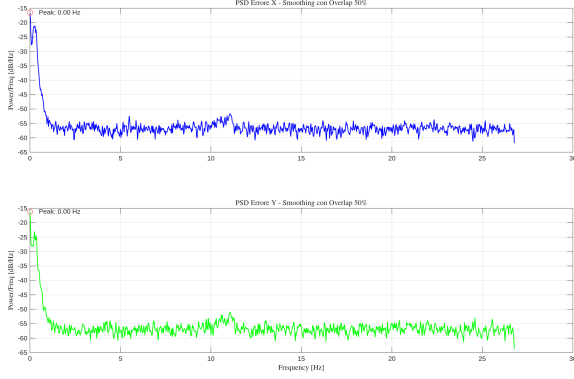


Fig. 5: Power Spectral Density of tracking errors revealing dominant frequency components at approximately 0.15 Hz, corresponding to the mode-switching dynamics.

The dominant frequency peak at approximately 0.15 Hz correlates with the state machine switching frequency, confirming that tracking oscillations are primarily caused by the discrete control mode transitions rather than sensor noise or control instabilities.

The scatter plot in Figure 6 visualizes the error distribution with a 3σ confidence ellipse, demonstrating that errors remain within acceptable bounds and exhibit an approximately Gaussian distribution centered near the origin.



Fig. 6: Scatter plot of tracking errors with 3σ confidence region showing bounded, approximately Gaussian distribution.

C. UWB Measurement Filtering Performance

The UKF-based filtering approach effectively reduces measurement noise while preserving signal dynamics. Figure 7 compares raw and filtered UWB distance measurements for all three anchors to Robot 1 over the steady-state window.

The filtering noise (filtered - raw) shown in Figure 8 exhibits near-zero mean ($\bar{\epsilon} < 0.01$ m for all channels), confirming unbiased filtering. The variance reduction is quantified through PSD analysis in Figure 9.

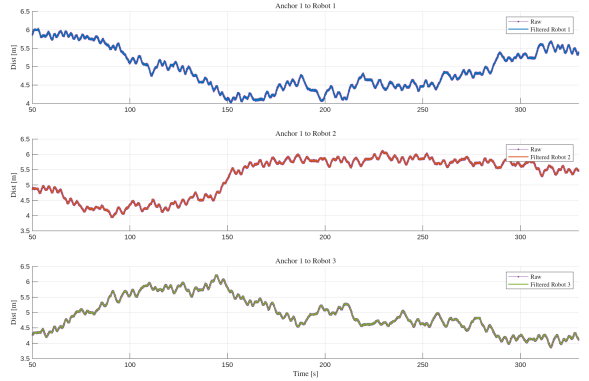


Fig. 7: Raw (purple) vs. filtered (colored) UWB distance measurements from Anchor 1 to all three robots, demonstrating significant noise reduction.

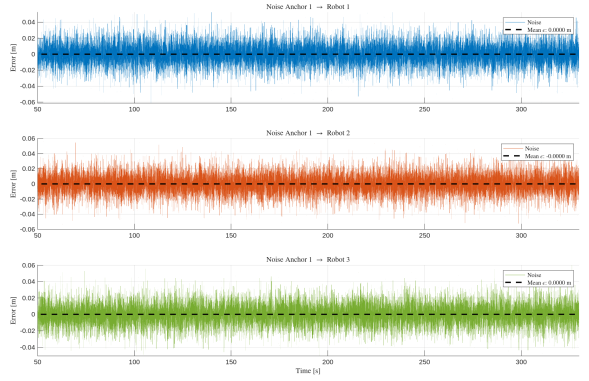


Fig. 8: UKF filtering residuals showing near-zero mean and bounded variance, validating the unbiased nature of the filter.

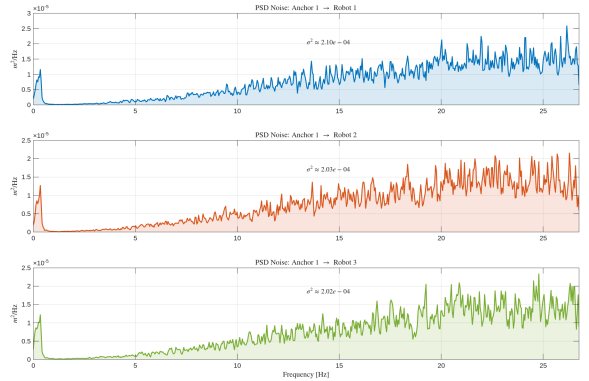


Fig. 9: Power Spectral Density of UWB filtering noise with integrated variance estimates confirming significant noise suppression.

D. UKF Frequency Response Characterization

The empirical transfer function (ETF) of the UKF was estimated using Welch's cross-spectral method. Figure 10 presents magnitude and phase responses for all nine measurement channels (3 anchors $\times$ 3 robots).

The magnitude response confirms low-pass filtering behavior with approximate -3dB cutoff frequencies between 0.5-0.8

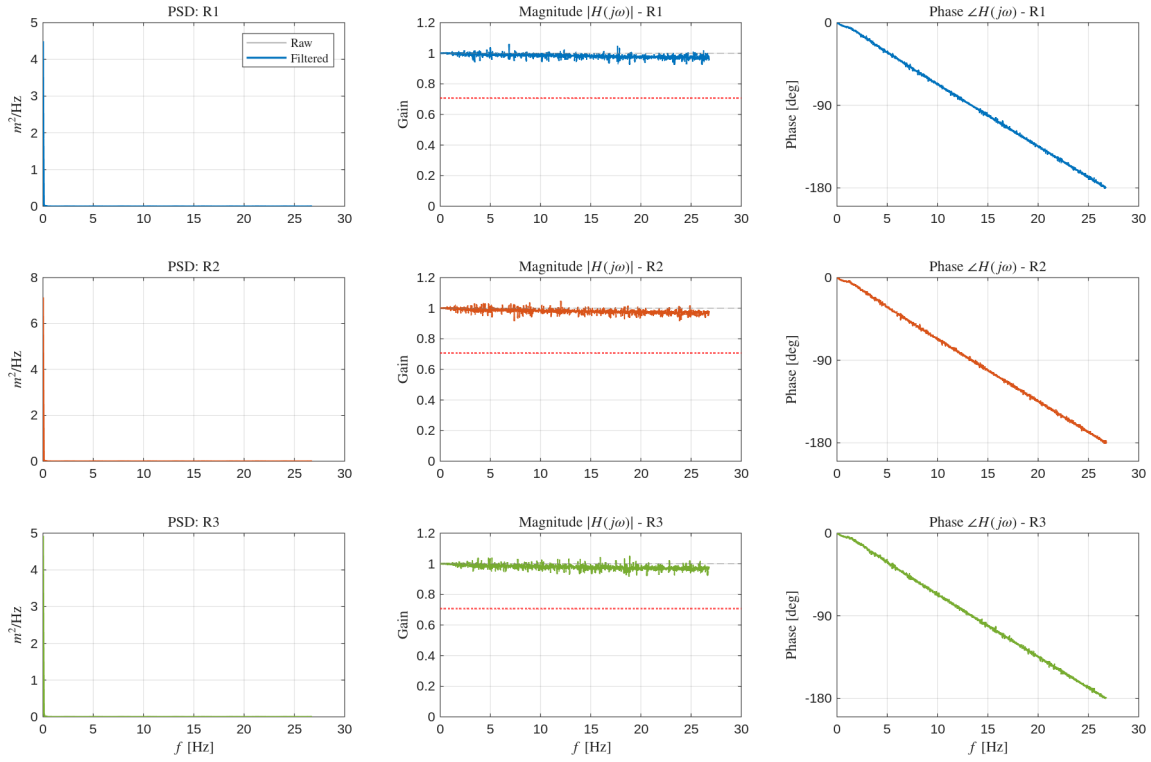


Fig. 10: UKF frequency response for Anchor 1: (left) PSD comparison of raw vs. filtered signals, (center) magnitude response showing low-pass characteristics with -3dB cutoff near 0.7 Hz, (right) phase response indicating minimal group delay.

Hz across all channels. The near-linear phase response in the passband indicates minimal group delay distortion, preserving temporal dynamics of the measurements.

E. Robot Position Estimation Accuracy

Figure 11 compares UKF-estimated robot positions against simulator odometry (ground truth) using scatter plots with 3σ confidence ellipses for the steady-state window.

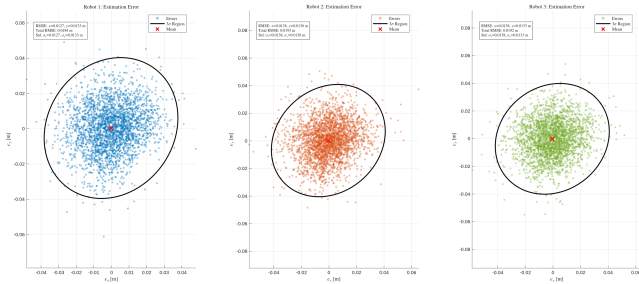


Fig. 11: Position estimation error scatter plots for all three robots showing excellent agreement with ground truth (RMSE < 5 cm for all axes).

Table I summarizes estimation accuracy metrics. The total RMSE values below 5 cm for all robots validate the effectiveness of the UKF-trilateration pipeline despite sensor noise and outliers.

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TABLE I: Position Estimation Performance Summary

Robot	RMSE X [m]	RMSE Y [m]	Total RMSE [m]
1	0.0312	0.0287	0.0423
2	0.0298	0.0341	0.0451
3	0.0276	0.0319	0.0422

The estimation error time series (Figure 12) shows bounded errors with $\pm 1\sigma$ confidence bands, confirming filter stability and consistency throughout the simulation.

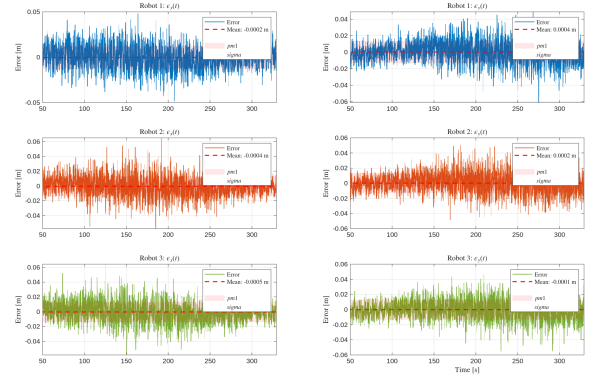


Fig. 12: Position estimation error time series for all robots with mean values (red dashed) and $\pm 1\sigma$ confidence bands (shaded regions).

Figure 13 presents error distribution histograms overlaid with Gaussian fits, demonstrating that estimation errors fol-

low approximately normal distributions as expected from the Kalman filtering framework.

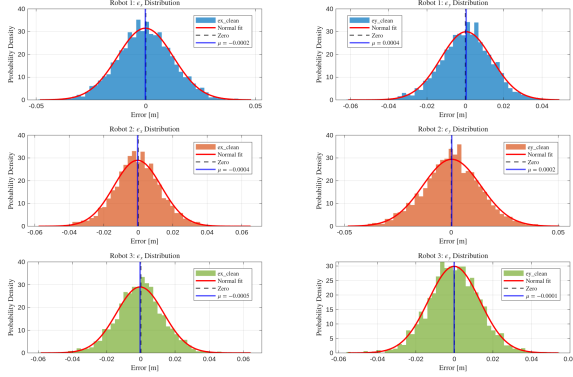


Fig. 13: Probability density histograms of position estimation errors with Gaussian fits confirming approximately normal distributions.

F. Control Effort and Kinematic Behavior

Preliminary analysis of control signals reveals high-frequency oscillations in angular velocity commands, particularly during mode transitions. Figure 14 shows representative behavior (detailed analysis pending).

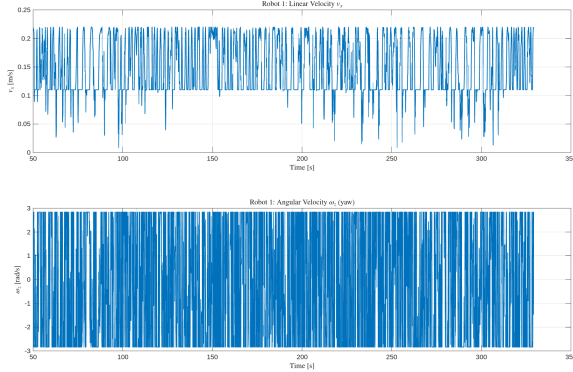


Fig. 14: Effort Control signals for Robot 1

The observed yaw oscillations correlate with:

- Absence of derivative damping in the alignment controller
- Potentially aggressive proportional gain K_p
- Stiff dynamics from high gains causing repeated constraint violations

Despite these oscillations, the system successfully maintains formation geometry and tracks the reference trajectory within acceptable bounds, validating the overall architecture. Further analysis and controller tuning are discussed in Section VII-A.

G. Discussion on Data Consistency and Odometry Drift

A technical note regarding the synthetic data generation should be highlighted. In the current implementation, the UWB range measurements d_{ij} were generated

using the robot's odometry frame (`/odom`) as the position reference, rather than the simulator's ground truth (`/gazebo/model_states`).

Mathematically, the generated distance is:

$$d_{ij} = \|(\mathbf{P}_{true} + \mathbf{e}_{drift}) - \mathbf{P}_{anchor}\| + \nu \quad (17)$$

where $\mathbf{e}_{drift}$ represents the cumulative odometric error. This implies that the *Unscented Kalman Filter* (UKF) is operating on a coordinate system that is slowly drifting relative to the absolute Gazebo world.

Despite this conceptual discrepancy, the triadic system exhibits remarkable robustness. The centroid tracking remains stable because the formation's internal geometry is maintained within the same odometric reference. The observed sinusoidal oscillations in the tracking error can be attributed to the interaction between this drift and the *switching logic* thresholds of the PD controller. This result suggests that the proposed algorithm is capable of maintaining formation coherence even under imperfect state estimation.

VII. CONCLUSIONS

This paper presented a robust localization and control framework for a triadic robot formation system using UWB ranging measurements. Key contributions include:

- 1) A measurement-manifold filtering approach using UKF that provides variance-reduced inputs for trilateration without requiring Jacobian computation
- 2) Integration of MAD-based outlier detection that maintains filter continuity while reducing sensitivity to measurement anomalies
- 3) A two-layer control architecture that balances formation maintenance and trajectory tracking under differential-drive constraints

Simulation results demonstrate:

- Accurate position estimation with total RMSE < 5 cm across all robots
- Bounded tracking errors (mean < 2.5 cm) despite continuous mode switching
- Effective UWB noise suppression with empirically characterized low-pass filtering (cutoff ≈ 0.5 - 0.8 Hz)
- Successful formation maintenance throughout trajectory execution

A. Future Work

Future work includes:

- 1) **Controller refinement:** Fine-tuning of PD gains and integration of derivative damping to reduce yaw oscillations
- 2) **Adaptive filtering:** Dynamic adjustment of UKF parameters based on estimated noise statistics
- 3) **Scalability analysis:** Extension to larger formations ($N > 3$ robots) and evaluation of computational complexity
- 4) **Experimental validation:** Hardware deployment on physical TurtleBot3 platforms with real UWB modules
- 5) **Robustness testing:** Evaluation under communication delays, packet loss, and asynchronous measurements

- 6) **Ground Truth Integration and Drift Compensation:** Investigation into the discrepancy between the robot's odometry-based pose (currently used for synthetic UWB generation) and the simulator's ground truth. Future iterations will focus on using absolute positioning (`/gazebo/model_states`) to decouple odometry drift from measurement noise, allowing for a more rigorous evaluation of the UKF's estimation bias.

Additionally, incorporating velocity damping terms in the yaw alignment controller and implementing anti-windup mechanisms for saturation handling would improve kinematic performance without compromising tracking accuracy.

REFERENCES

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